

# Supplement: Next-Day Municipal Service-Demand Forecasting Across Four Cities

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## 1 Extended Notes on Data Provenance

### Sources and licenses.

- **New York:** *311 Service Requests* (current dataset, scoped 2020 to present — which sets the benchmark’s uniform 2020-01-01 window start, decision D14), NYC Open Data, dataset `erm2-nwe9`; NYC Open Data terms (free use without restriction).
- **Chicago:** *311 Service Requests*, Chicago Data Portal, dataset `v6vf-nfxy`; covers the current 311 system live since 2018-12-18 (the uniform study window starts 2020-01-01 per decision D14, well clear of the system transition); source-flagged duplicates dropped.
- **San Francisco:** *311 Cases*, DataSF, dataset `vw6y-z8j6`; Public Domain Dedication and License.
- **Austin:** *Austin 311 Public Data*, City of Austin Open Data Portal, dataset `xwdj-i9he`.

- **Weather:** NOAA GHCN-Daily through the AWS Open Data Program mirror `noaa-ghcn-pds` (maintained by NOAA NCEI); stations USW00094728 (Central Park), USW00094846 (O’Hare), USW00023272 (San Francisco Downtown), USW00013904 (Camp Mabry). Native units (tenths of mm/°C, tenths of m/s) converted only in preprocessing.

**Acquisition queries.** Each 311 source is queried through the SODA API with server-side aggregation — `$select=date_trunc_ymd(created_date), category, count(*)`, grouped and ordered, chunked by calendar year, paginated at 50,000 rows — so the rawest stored layer is already free of record-level attributes. The complete list of issued query URLs for every city is in `data/raw/311/*_manifest.json`.

**Known data characteristics.** Reported request volumes reflect intake-channel structure (phone/app/web), administrative practice, and reporting propensity, all of which differ across cities and over time; the Chicago information-only and aircraft-noise exclusions and the SF duplicate exclusion are documented with reasons in `configs/service_families.yml`. A missing (day, family) pair within a city’s observed range denotes zero requests created, and the panel densification step materializes those zeros explicitly.

## 2 Service-Family Harmonization

The harmonization rule set (`configs/service_families.yml`) is an ordered, case-insensitive substring matcher: the first matching family wins, and categories matching no rule fall to *other*. Order resolves genuine ambiguities deterministically (e.g., “noise” complaints about construction are noise, not housing/building, because the noise rule precedes). The full mapping from every native category to its family, with request counts, is exported per city to `outputs/tables/family_composition_{city}.csv` by the preprocessing stage and ships with the repository; the tables below summarize the dominant native categories per family in each city.

Two properties matter for interpretation. First, the rules are *versioned*: any change to the mapping is a visible diff, and the unit test suite pins representative mappings. Second, family compositions are *not* identical across cities — e.g., the noise family is dominated by residential-noise complaints in New York but is a much smaller share elsewhere; the sanitation family includes encampment-related categories in San Francisco that have no Chicago counterpart. All cross-city statements in the main text are therefore about families as *planning buckets within each city*, and the cross-city axis compares the structure of the accuracy-to-decision mapping, never raw family volumes.

### Austin

- **housing building** (1.9%): DSD - Outdoor Commercial Venue Music Complaint (7,867); Park Maintenance - Grounds Plumbing Issues (3,529); ATD - Construction Items in Right of Way (3,132)
- **noise** (4.7%): APD - Non Emergency Noise/Alarm (70,197)
- **other** (23.5%): Loose Dog (27,062); 311 CC - Other (25,843); Animal Protection - Loose Dog (24,018)
- **parks trees** (4.4%): Park Maintenance - Grounds (37,444); Tree Issue ROW (13,396); PWD - Tree Issue Right of Way (5,068)

- **public safety** (3.2%): APD - Non Emergency Collision (32,924); Sign - Traffic Sign Emergency (2,185); ZZ - AFD - Fireworks Complaint (2,049)
- **sanitation** (21.3%): ARR - Garbage (79,958); ARR - Compost (49,879); ARR - Storm Debris Collection (49,751)
- **streets transport** (38.5%): Austin Code - Request Code Officer (81,975); DSD - Request Code Officer (47,058); ATD - Parking Violation Enforcement (42,161)
- **water sewer** (2.6%): AW - Water Conservation Violation (10,628); WPD - Channels/Creek/Drainage Issues (7,643); AW - Water Waste Report (3,978)

## Chicago

- **housing building** (6.9%): Building Violation (107,110); Water Lead Test Kit Request (65,635); Buildings - Plumbing Violation (40,969)
- **other** (6.9%): Stray Animal Complaint (61,065); Report an Injured Animal (50,921); Vicious Animal Complaint (37,396)
- **parks trees** (10.3%): Tree Trim Request (NO LONGER BEING ACCEPTED) (206,525); Weed Removal Request (108,205); Tree Removal Inspection (96,345)
- **public safety** (2.6%): Tree Emergency (121,023); Commercial Fire Safety Inspection Request (1,557)
- **sanitation** (38.3%): Graffiti Removal Request (556,086); Rodent Baiting/Rat Complaint (295,754); Garbage Cart Maintenance (282,964)
- **streets transport** (28.7%): Abandoned Vehicle Complaint (269,189); Pothole in Street Complaint (215,010); Street Light Out Complaint (197,946)
- **water sewer** (6.3%): Water in Basement Complaint (56,251); Sewer Cleaning Inspection Request (52,303); Check for Leak (45,813)

## New York

- **housing building** (20.5%): HEAT/HOT WATER (1,439,229); UNSANITARY CONDITION (605,801); PLUMBING (343,419)
- **noise** (23.3%): Noise - Residential (2,287,530); Noise - Street/Sidewalk (1,027,860); Noise - Vehicle (374,275)
- **other** (6.2%): GENERAL (166,982); Consumer Complaint (138,005); Maintenance or Facility (121,806)
- **parks trees** (2.9%): Damaged Tree (214,620); Overgrown Tree/Branches (115,202); New Tree Request (93,700)
- **public safety** (2.7%): Non-Emergency Police Matter (190,623); Illegal Fireworks (112,922); Drug Activity (86,992)
- **sanitation** (11.5%): Request Large Bulky Item Collection (632,148); Dirty Condition (224,713); Rodent (220,299)

- **streets transport** (28.5%): Illegal Parking (2,478,000); Blocked Driveway (934,597); Street Condition (410,510)
- **water sewer** (4.4%): Water System (366,096); WATER LEAK (220,757); Sewer (172,464)

## San Francisco

- **housing building** (0.6%): Residential Building Request (11,712); General Request - BUILDING INSPECTION (7,468); Residential Building (6,364)
- **noise** (1.3%): Noise Report (37,785); Noise (18,133)
- **other** (8.8%): General Request - PUBLIC WORKS (105,723); General Request (62,711); General Request - MTA (54,573)
- **parks trees** (2.8%): Tree Maintenance (80,947); Rec and Park Requests (40,539)
- **public safety** (0.0%): General Request - FIRE DEPARTMENT (490)
- **sanitation** (19.6%): Graffiti (337,617); Encampments (158,952); Graffiti Public (121,575)
- **streets transport** (65.4%): Street and Sidewalk Cleaning (1,933,456); Parking Enforcement (602,068); Abandoned Vehicle (62,042)
- **water sewer** (1.6%): Sewer Issues (47,683); Sewer (18,406); Catch Basin Maintenance (2,161)

## 3 Model Configurations

Hyperparameters are fixed across cities and feature sets so that no configuration receives a larger tuning budget than another; values were chosen a priori in the ranges standard for daily-count panels of this size and are not per-task optimized.

| Model             | Parameters                                                               | Notes                            |
|-------------------|--------------------------------------------------------------------------|----------------------------------|
| Trailing mean     | 7-day window                                                             | forecast-free baseline           |
| Seasonal naive    | same weekday, $t - 6$                                                    | forecast-free baseline           |
| Ridge             | $\alpha = 1.0$ , standardized inputs                                     |                                  |
| Random forest     | 300 trees, min leaf 5, 60% features                                      | <a href="#">Breiman (2001)</a>   |
| LightGBM point    | L1 objective, 600 trees, lr 0.05, 63 leaves, min child 30                | <a href="#">Ke et al. (2017)</a> |
| LightGBM quantile | as above, pinball objective per level<br>non-crossing via cumulative max | levels .05/.25/.50/.75/.95       |

As the target is a non-negative count, a Poisson GLM (log link, standardized inputs) is reported as an additional count-data baseline (`tab21_poisson`); fit per city and feature set on train+validation and evaluated once on test, it beats the naive trailing-mean in 11 of 12 (city, feature set) cells but is dominated by the validation-selected gradient-boosted and tree models in all 12, so it is reported but is not a candidate in the pre-registered selection or decision grids.

Categorical handling: one-hot family indicators everywhere; one-hot city indicators in pooled (global) models only. Predictions are clipped at zero. The selected configuration per (scope, city, feature set, model) is refit on train+validation before its single test evaluation, exactly as in the protocol of the main text.

## 4 Allocation Optimality

The greedy allocation used by all expected-value policies is exactly optimal for the per-day problem it solves.

**Proposition 1.** *Let  $D_1, \dots, D_S$  be non-negative random variables,  $\omega_s > 0$ ,  $\kappa > 0$ , and consider*

$$\max_{x \in \mathbb{Z}_{\geq 0}^S, \sum_s x_s = B} \sum_{s=1}^S \omega_s \mathbb{E}[\min(D_s, \kappa x_s)].$$

*The greedy algorithm that assigns capacity units one at a time to the family with the largest marginal gain attains the optimum.*

*Proof.* For fixed  $s$ , define  $g_s(x) = \omega_s \mathbb{E}[\min(D_s, \kappa x)]$  on integers  $x \geq 0$ . For any realization  $d \geq 0$  the map  $c \mapsto \min(d, c)$  is concave in  $c$ ; expectation preserves concavity, and restriction to the integer grid yields non-increasing increments  $\Delta_s(x) = g_s(x+1) - g_s(x)$ . The problem is therefore the maximization of a separable discrete-concave function over the simplex  $\sum_s x_s = B$ , for which the greedy/incremental algorithm is optimal: at every step the chosen increment is at least as large as any increment available later (by monotonicity of  $\Delta_s$ ), so an exchange argument converts any optimal solution into the greedy one without loss of objective value.  $\square$

In the implementation  $\mathbb{E}[\min(D_s, c)]$  is evaluated exactly for the discrete distribution defined by linear interpolation of the inverse CDF through the five forecast quantiles on a fixed 99-point grid (`src/optimization/allocation.py`); point forecasts use a degenerate distribution, and the hind-sight myopic reference uses the realized value. The unit test `test_greedy_matches_bruteforce_on_small_instances` verifies greedy-vs-brute-force equality on random instances.

**The piecewise-flat marginal-value profile of degenerate forecasts (mechanism, both segments).** For a degenerate distribution at point value  $d$ ,  $\mathbb{E}[\min(D, \kappa x)] = \min(d, \kappa x)$ , so the marginal value of each additional unit equals  $\kappa\omega$  while  $\kappa x < d$  (the deficit segment) and 0 once  $\kappa x \geq d$  (the surplus segment). Under a neutral objective the greedy ordering therefore carries no information in *either* segment, and allocation is decided by tie-breaking. Worked examples (computed with the repository’s own `greedy_allocate`): *surplus* — budget 12, three families with point demand 30 each,  $\kappa = 10$ : greedy returns  $[6, 3, 3]$ , piling the surplus uninformatively; *deficit* — budget 4 with the same demands: greedy returns  $[3, 1, 0]$  versus proportional  $[2, 1, 1]$ ; with unequal demands 50/30/10 and budget 4, greedy returns  $[4, 0, 0]$  versus proportional  $[2, 1, 1]$ . Distributional tails make  $\mathbb{E}[\min(D, \kappa x)]$  strictly concave and restore an informative gradient in both segments.

## 5 Full Result Tables

All tables regenerate from `scripts/make_tables_figures.py`; CSV versions live in `outputs/tables/` with provenance hashes in `_provenance.json`.

Validation-selected headline (test MAE of the validation-chosen model per city and feature set):

| city    | internal | calendar | weather | calendar_weather |
|---------|----------|----------|---------|------------------|
| austin  | 16.529   | 14.843   | 16.530  | 14.844           |
| chicago | 60.374   | 50.945   | 58.665  | 49.295           |
| nyc     | 207.022  | 182.378  | 186.728 | 165.417          |
| sf      | 35.183   | 31.532   | 35.866  | 30.061           |

Selection detail (chosen model, validation MAE, single test evaluation):

| city    | feature_set                  | selected_model | val_mae | test_mae |
|---------|------------------------------|----------------|---------|----------|
| austin  | calendar                     | lgbm_point     | 15.389  | 14.843   |
| austin  | calendar_weather             | lgbm_point     | 15.245  | 14.844   |
| austin  | calendar_weather_lagged_only | lgbm_point     | 15.711  | 14.865   |
| austin  | internal                     | random_forest  | 16.494  | 16.529   |
| austin  | weather                      | lgbm_point     | 16.462  | 16.530   |
| chicago | calendar                     | lgbm_point     | 48.527  | 50.945   |
| chicago | calendar_weather             | lgbm_point     | 46.802  | 49.295   |
| chicago | calendar_weather_lagged_only | lgbm_point     | 47.997  | 50.388   |
| chicago | internal                     | random_forest  | 54.803  | 60.374   |
| chicago | weather                      | lgbm_point     | 52.580  | 58.665   |
| nyc     | calendar                     | lgbm_point     | 146.497 | 182.378  |
| nyc     | calendar_weather             | lgbm_point     | 131.660 | 165.417  |
| nyc     | calendar_weather_lagged_only | lgbm_point     | 150.323 | 178.626  |
| nyc     | internal                     | lgbm_point     | 165.701 | 207.022  |
| nyc     | weather                      | lgbm_point     | 153.963 | 186.728  |
| sf      | calendar                     | lgbm_point     | 30.341  | 31.532   |
| sf      | calendar_weather             | lgbm_point     | 28.973  | 30.061   |
| sf      | calendar_weather_lagged_only | lgbm_point     | 30.173  | 31.659   |
| sf      | internal                     | ridge          | 31.435  | 35.183   |
| sf      | weather                      | ridge          | 31.850  | 35.866   |

Exhaustive descriptive grid (every model  $\times$  feature set; never used for selection):

| city    | model           | internal | calendar | weather | calendar_weather |
|---------|-----------------|----------|----------|---------|------------------|
| austin  | lgbm_point      | 16.719   | 14.843   | 16.530  | 14.844           |
| austin  | naive_trailing7 | 30.371   | 30.371   | 30.371  | 30.371           |
| austin  | random_forest   | 16.529   | 15.506   | 16.617  | 15.467           |
| austin  | ridge           | 24.976   | 25.677   | 24.955  | 25.609           |
| austin  | seasonal_naive7 | 21.613   | 21.613   | 21.613  | 21.613           |
| chicago | lgbm_point      | 59.236   | 50.945   | 58.665  | 49.295           |
| chicago | naive_trailing7 | 111.461  | 111.461  | 111.461 | 111.461          |
| chicago | random_forest   | 60.374   | 54.418   | 59.268  | 53.741           |
| chicago | ridge           | 71.809   | 70.510   | 72.851  | 70.810           |
| chicago | seasonal_naive7 | 84.238   | 84.238   | 84.238  | 84.238           |
| nyc     | lgbm_point      | 207.022  | 182.378  | 186.728 | 165.417          |
| nyc     | naive_trailing7 | 305.618  | 305.618  | 305.618 | 305.618          |
| nyc     | random_forest   | 209.281  | 195.088  | 200.152 | 190.629          |
| nyc     | ridge           | 215.154  | 213.486  | 214.625 | 213.113          |
| nyc     | seasonal_naive7 | 282.439  | 282.439  | 282.439 | 282.439          |
| sf      | lgbm_point      | 35.046   | 31.532   | 33.225  | 30.061           |
| sf      | naive_trailing7 | 44.769   | 44.769   | 44.769  | 44.769           |
| sf      | random_forest   | 34.515   | 32.840   | 33.215  | 31.562           |
| sf      | ridge           | 35.183   | 36.562   | 35.866  | 36.691           |
| sf      | seasonal_naive7 | 43.786   | 43.786   | 43.786  | 43.786           |

Quantile-forecast evaluation (one unremarked pattern worth noting: the pooled quantile model’s 90% test coverage, 0.745–0.872, is closer to nominal than the local model’s in three of four cities, even though the local model is the decision layer’s input):

| scope  | city    | model         | pinball | coverage_90 | coverage_50 | mae_median |
|--------|---------|---------------|---------|-------------|-------------|------------|
| local  | austin  | lgbm_quantile | 5.162   | 0.812       | 0.413       | 14.790     |
| local  | chicago | lgbm_quantile | 17.285  | 0.768       | 0.375       | 48.959     |
| local  | nyc     | lgbm_quantile | 58.980  | 0.769       | 0.342       | 167.194    |
| local  | sf      | lgbm_quantile | 10.189  | 0.808       | 0.344       | 29.286     |
| global | austin  | lgbm_quantile | 4.991   | 0.872       | 0.485       | 15.169     |
| global | chicago | lgbm_quantile | 17.618  | 0.831       | 0.456       | 52.561     |
| global | nyc     | lgbm_quantile | 58.862  | 0.818       | 0.408       | 170.492    |
| global | sf      | lgbm_quantile | 9.748   | 0.745       | 0.399       | 29.611     |

Rank agreement between accuracy and simulated-decision orderings (descriptive):

| city    | regime   | n_models | spearman_rho |
|---------|----------|----------|--------------|
| austin  | generous | 11       | 0.991        |
| austin  | moderate | 11       | 0.991        |
| austin  | scarce   | 11       | 0.982        |
| chicago | generous | 11       | 0.955        |
| chicago | moderate | 11       | 0.936        |
| chicago | scarce   | 11       | 0.955        |
| nyc     | generous | 11       | 0.964        |
| nyc     | moderate | 11       | 0.964        |
| nyc     | scarce   | 11       | 0.964        |
| sf      | generous | 11       | 0.882        |
| sf      | moderate | 11       | 0.900        |
| sf      | scarce   | 11       | 0.900        |

Validation selection by MAE versus by decision loss (frozen budgets):

| city    | regime   | selected_by_val_mae                | selected_by_val_decision           | test_loss_mae_choice |
|---------|----------|------------------------------------|------------------------------------|----------------------|
| austin  | scarce   | global/calendar_weather/lgbm_point | local/calendar_weather/lgbm_point  | 16572872.00          |
| austin  | moderate | global/calendar_weather/lgbm_point | local/calendar_weather/lgbm_point  | 10869184.00          |
| austin  | generous | global/calendar_weather/lgbm_point | local/calendar_weather/lgbm_point  | 7085533.00           |
| chicago | scarce   | local/calendar_weather/lgbm_point  | local/calendar_weather/lgbm_point  | 38101878.00          |
| chicago | moderate | local/calendar_weather/lgbm_point  | local/calendar_weather/lgbm_point  | 21402749.00          |
| chicago | generous | local/calendar_weather/lgbm_point  | local/calendar_weather/lgbm_point  | 6920807.00           |
| nyc     | scarce   | local/calendar_weather/lgbm_point  | local/calendar_weather/lgbm_point  | 154437576.00         |
| nyc     | moderate | local/calendar_weather/lgbm_point  | local/calendar_weather/lgbm_point  | 91571785.00          |
| nyc     | generous | local/calendar_weather/lgbm_point  | local/calendar_weather/lgbm_point  | 27875499.00          |
| sf      | scarce   | global/calendar_weather/lgbm_point | global/calendar_weather/lgbm_point | 30255010.00          |
| sf      | moderate | global/calendar_weather/lgbm_point | global/calendar_weather/lgbm_point | 16972510.00          |
| sf      | generous | global/calendar_weather/lgbm_point | global/calendar_weather/lgbm_point | 3952067.00           |

The complete decision grid (including the labeled, non-bounding hindsight myopic diagnostic) is `tab3_decision_all.csv`; block-bootstrap intervals for the pre-named contrasts are `tab8_inference.csv` (decision) and `tab9_forecast_inference.csv` (forecast), each at block lengths 14/28/56.

**Dataset audit.** Acquired, excluded, and retained records and native-category counts per city (from the build manifests):

| City          | Dataset id | Retrieved  | Native cats | Acquired | Excluded | Retained |
|---------------|------------|------------|-------------|----------|----------|----------|
| New York      | erm2-nwe9  | 2026-06-09 | 271         | 19669371 | 0        | 19669371 |
| Chicago       | v6vf-nfxv  | 2026-06-09 | 107         | 11309296 | 6625786  | 4683510  |
| San Francisco | vw6y-z8j6  | 2026-06-09 | 136         | 4374676  | 0        | 4374676  |
| Austin        | xwdj-i9he  | 2026-06-09 | 346         | 1487588  | 0        | 1487588  |
| Total         |            |            | 860         | 36840931 | 6625786  | 30215145 |

**Chronological split boundaries.** Exact per-city train / validation / test day ranges (the modeling window ends at the last day of acquired data; the densified panel carries the nominal study window to 2025-12-31):

| City          | Train                    | Validation               | Test                     |
|---------------|--------------------------|--------------------------|--------------------------|
| Austin        | 2020-01-28 .. 2023-08-04 | 2023-08-05 .. 2024-05-05 | 2024-05-06 .. 2025-02-05 |
| Chicago       | 2020-01-28 .. 2023-08-03 | 2023-08-04 .. 2024-05-04 | 2024-05-05 .. 2025-02-04 |
| New York      | 2020-01-28 .. 2023-08-03 | 2023-08-04 .. 2024-05-04 | 2024-05-05 .. 2025-02-04 |
| San Francisco | 2020-01-28 .. 2023-07-29 | 2023-07-30 .. 2024-05-06 | 2024-05-07 .. 2025-02-05 |

**Eleven-configuration cross-scope grid.** The pre-specified candidate set for the Section 7 selection experiment is, exactly: ten local configurations — {naive trailing-7, seasonal-naive-7, ridge, random forest, LightGBM}×internal; {random forest, LightGBM}×calendar; {ridge, random forest, LightGBM}×calendar+weather — plus the stage-censored global LightGBM on calendar+weather. No quantile model is a candidate.

**Objective non-identification (full grid).** The compact main-text view is Table ??; the full grid of every tie-break rule and control policy across all 12 city-regime cells is `tab12.tiebreak_full.csv`, and the per-cell tie frequency is `tie_frequency.csv`.

**Conformal recalibration.** Empirical 90% coverage and mean interval width before and after split-conformal recalibration, and the calibrated-vs-uncalibrated full-arm decision losses:

| city    | coverage90_uncal | coverage90_conformal | width90_uncal | width90_conformal | delta_q05 | delta_q95 |
|---------|------------------|----------------------|---------------|-------------------|-----------|-----------|
| austin  | 0.812            | 0.903                | 56.217        | 69.526            | -3.468    | 9.845     |
| chicago | 0.768            | 0.899                | 172.383       | 207.667           | -11.618   | 23.928    |
| nyc     | 0.769            | 0.883                | 503.653       | 642.612           | -19.792   | 119.167   |
| sf      | 0.808            | 0.894                | 90.932        | 101.241           | -4.215    | 6.691     |

| city    | regime   | units | full_uncal_loss | full_conformal_loss | proportional_loss | conformal_vs_uncal_pct | conformal_vs. |
|---------|----------|-------|-----------------|---------------------|-------------------|------------------------|---------------|
| austin  | scarce   | 8     | 16410288.000    | 16398015.000        | 16383102.000      | -0.075                 |               |
| austin  | moderate | 11    | 10679297.000    | 10667024.000        | 10651410.000      | -0.115                 |               |
| austin  | generous | 13    | 6856849.000     | 6844424.000         | 6838453.000       | -0.181                 |               |
| chicago | scarce   | 29    | 37208413.000    | 37183378.000        | 37162683.000      | -0.067                 |               |
| chicago | moderate | 38    | 20087932.000    | 20040119.000        | 19965951.000      | -0.238                 |               |
| chicago | generous | 46    | 5561653.000     | 5546422.000         | 5451400.000       | -0.274                 |               |
| nyc     | scarce   | 119   | 146938634.000   | 146813066.000       | 146024681.000     | -0.085                 |               |
| nyc     | moderate | 153   | 82149267.000    | 81984117.000        | 81040481.000      | -0.201                 |               |
| nyc     | generous | 187   | 18748666.000    | 18583257.000        | 17595851.000      | -0.882                 |               |
| sf      | scarce   | 26    | 29644860.000    | 29642224.000        | 29492173.000      | -0.009                 |               |
| sf      | moderate | 33    | 16362360.000    | 16359724.000        | 16204987.000      | -0.016                 |               |
| sf      | generous | 41    | 3693238.000     | 3690775.000         | 3733466.000       | -0.067                 |               |

**Normalized pooling.** Pooled-minus-local test MAE (% of local) under raw, log1p, and per-city-standardized targets:

| city    | pooled_log1p_pct_vs_local | pooled_per_city_z_pct_vs_local | pooled_raw_pct_vs_local |
|---------|---------------------------|--------------------------------|-------------------------|
| austin  | 1.980                     | 2.030                          | 4.100                   |
| chicago | 5.050                     | 4.020                          | 6.500                   |
| nyc     | 2.100                     | 1.560                          | 1.800                   |
| sf      | -0.560                    | -2.410                         | -1.170                  |



**Horizon robustness.** Policy ranking at the first 30/60/90/180 and full test days is `tab15_horizon.csv`; the claim-bearing orderings (uniform worst; fixed-index point-greedy below all identified policies) are stable across horizons, while the three near-tied identified policies reshuffle within  $< 1\%$ .

**Per-family served fraction.** Served fraction by family under the uniform floor and the full-distribution arm (moderate regime), so any systematic per-family deprioritization is visible:

| city    | family            | quantile/full_arm | uniform |
|---------|-------------------|-------------------|---------|
| austin  | housing_building  | 0.997             | 1.000   |
| austin  | noise             | 0.997             | 1.000   |
| austin  | other             | 0.999             | 0.589   |
| austin  | parks_trees       | 0.993             | 1.000   |
| austin  | public_safety     | 0.998             | 1.000   |
| austin  | sanitation        | 0.997             | 0.291   |
| austin  | streets_transport | 0.186             | 0.183   |
| austin  | water_sewer       | 0.000             | 1.000   |
| chicago | housing_building  | 0.998             | 1.000   |
| chicago | other             | 0.998             | 1.000   |
| chicago | parks_trees       | 0.997             | 0.998   |
| chicago | public_safety     | 0.999             | 1.000   |
| chicago | sanitation        | 1.000             | 0.286   |
| chicago | streets_transport | 0.673             | 0.426   |
| chicago | water_sewer       | 0.000             | 1.000   |
| nyc     | housing_building  | 0.999             | 0.455   |
| nyc     | noise             | 1.000             | 0.402   |
| nyc     | other             | 0.999             | 1.000   |
| nyc     | parks_trees       | 1.000             | 1.000   |
| nyc     | public_safety     | 0.999             | 1.000   |
| nyc     | sanitation        | 0.982             | 1.000   |
| nyc     | streets_transport | 0.356             | 0.326   |
| nyc     | water_sewer       | 0.003             | 1.000   |
| sf      | housing_building  | 0.990             | 1.000   |
| sf      | noise             | 0.998             | 1.000   |
| sf      | other             | 1.000             | 1.000   |
| sf      | parks_trees       | 0.995             | 1.000   |
| sf      | public_safety     | 0.000             | 1.000   |
| sf      | sanitation        | 1.000             | 0.491   |
| sf      | streets_transport | 0.686             | 0.136   |
| sf      | water_sewer       | 0.000             | 1.000   |

**Poisson count baseline.** Naive, Poisson GLM, and the validation-selected model (test MAE) per city and feature set:

| city    | feature_set      | naive_mae | poisson_glm_mae | selected_model | selected_mae |
|---------|------------------|-----------|-----------------|----------------|--------------|
| austin  | internal         | 30.371    | 31.228          | random_forest  | 16.529       |
| austin  | calendar         | 30.371    | 26.055          | lgbm_point     | 14.843       |
| austin  | calendar_weather | 30.371    | 26.126          | lgbm_point     | 14.844       |
| chicago | internal         | 111.461   | 90.160          | random_forest  | 60.374       |
| chicago | calendar         | 111.461   | 74.126          | lgbm_point     | 50.945       |
| chicago | calendar_weather | 111.461   | 73.525          | lgbm_point     | 49.295       |
| nyc     | internal         | 305.618   | 263.048         | lgbm_point     | 207.022      |
| nyc     | calendar         | 305.618   | 253.983         | lgbm_point     | 182.378      |
| nyc     | calendar_weather | 305.618   | 253.053         | lgbm_point     | 165.417      |
| sf      | internal         | 44.769    | 41.869          | ridge          | 35.183       |
| sf      | calendar         | 44.769    | 36.664          | lgbm_point     | 31.532       |
| sf      | calendar_weather | 44.769    | 36.336          | lgbm_point     | 30.061       |

**Weather missingness.** Missing values after quality-flag drop and  $\leq 7$ -day interpolation, per city:

| City          | TMAX missing | TMIN missing | PRCP missing |
|---------------|--------------|--------------|--------------|
| New York      | 0            | 0            | 0            |
| Chicago       | 0            | 0            | 0            |
| San Francisco | 6            | 7            | 0            |
| Austin        | 0            | 0            | 0            |

**Guard suite.** What each protocol guard checks and against which artifact:

| Guard | Checks                                                                        | Artifact                     |
|-------|-------------------------------------------------------------------------------|------------------------------|
| G1    | Budgets from training window only; invariant to test perturbation             | frozen_budgets.json          |
| G2    | Model selection uses validation MAE only; invariant to corrupted test metrics | validation_selection.csv     |
| G3    | Leave-one-city-out transfer rows carry recomputed censor dates                | forecast_metrics.csv         |
| G4    | All three uncertainty arms derive from one fitted quantile model              | decision_metrics.csv         |
| G5    | Structural absence (Chicago noise) never encoded as zero rows                 | active_families.json         |
| G6    | Allocation conserves the full budget on uneven family sets                    | allocation.py                |
| G7/G8 | No forbidden terminology in output tables or manuscript                       | tables / *.tex               |
| G9    | Every sensitivity scenario in outputs matches the config grid                 | decision_sensitivity.csv     |
| G10   | Dataset ids and study window agree across config and manifests                | data_sources.yml / manifests |
| G11   | Provenance SHA-256 hashes bind every table/figure to the run                  | _provenance.json             |
| G12   | Simulated family sets equal the active-family manifest                        | decision_metrics.csv         |
| G13   | Pooled training is stage-censored (fit-time proof manifest)                   | pooled_censoring.json        |
| G14   | Printed dataset ids are a subset of the configured ids                        | *.tex                        |
| G15   | Table-1 caption carries the Chicago seven-family qualification                | data_stats.tex               |
| G16   | Cross-scope by-MAE selection equals the recomputed argmin                     | decision_selection.csv       |

## 6 Negative and Conditional Results

The corrected protocol produced several results that resolved against the more optimistic hypothesis; all are retained as findings:

- **Weather adds nothing in Austin:** +0.001 test MAE on top of calendar features (block-bootstrap interval covers zero), against a 9.3% gain in New York.
- **Pooled training hurts under valid censoring:** with stage-censored pooled fits (Amendment C11), pooling raises test MAE significantly in Austin (+4.1%), Chicago (+6.5%), and New York (+1.8%); San Francisco is neutral. Censored zero-shot transfer degrades MAE by +5% to +149%.
- **The exact point-forecast optimizer loses to proportional apportionment** in all 12 city-regime settings under the neutral objective — the identified mechanism being that a degenerate forecast makes the marginal-value profile piecewise flat in *both* segments ( $\kappa\omega$  below the point estimate plus carryover, zero above), so the greedy ordering is tie-broken rather than informed; distributional tails restore an allocation gradient (worked examples for both segments in the optimality section).
- **Quantile intervals are under-dispersed:** 90% nominal, 76.8–81.2% empirical; unrepaired in this paper.
- **A protocol defect was detected and corrected during review** (R1-F1): per-city split boundaries differ by 1–2 calendar days, so uncensored pooled fits contained  $\approx 0.1\%$  contemporaneous other-city rows. After the targeted censoring fix, the selection-experiment story

changed from 7/12 disagreements with 4 apparent harms to 9/12 agreements, 2 gains, 0 harms — the earlier disagreement was substantially an artifact of the contaminated pooled model’s inflated validation scores. The pre-fix outputs are preserved in the repository history for audit.

- **The hindsight myopic reference is not a bound:** being per-day myopic under carryover it can be exceeded; it appears only in the diagnostic table below and is never used as a denominator.

## 7 Sensitivity Analyses

Full numbers in `outputs/tables/tab7_sensitivity.csv`.

- **Normative-priority scenario** (illustrative, not socially validated): the proportional-optimizer gap compresses relative to the equal-weights primary; the main text therefore states the proportional-dominance claim as conditional on the neutral objective.
- **One-family-at-a-time service yield** ( $\times 0.70$  and  $\times 1.30$  for every active family, 1,125 sensitivity simulations): no policy-ordering reversal anywhere; the largest perturbation is New York, generous regime, noise yield  $\times 1.30$ .
- **10% daily abandonment:** policy ordering unchanged in all four cities.
- **Budget calibration on train+validation** instead of training only: conclusions unchanged.
- **Austin without the residual *other* family** (budgets recomputed train-only over the seven retained families): identical policy ordering to the primary analysis in all three regimes; Austin’s conclusions do not depend on its residual category.
- **Bootstrap block length** 14/28/56 days: 0 of 48 decision contrasts and none of the forecast contrasts change conclusion.
- **Trend diagnostic** (Newey–West lag-28 slope tests on the daily loss differentials; `tab10_trend_diagnostics`) 45 of 48 differential series and 11 of 12 loss level series are trend-dominated; only Chicago/generous is stationarity-compatible. Decision-contrast intervals are therefore read descriptively (main text, Sections 7–8). The diagnostic re-derives the daily series deterministically from the frozen inputs and asserts equality of every policy’s total loss with the E7 artifact to  $10^{-6}$  before computing any statistic. (San Francisco’s scarce and moderate rows coincide for the three within-greedy-family contrasts: at both budgets the greedy arms are fully saturated, so each serves exactly  $\kappa B$  per day and the paired differential depends only on the budget-invariant allocation-difference vector; the level series and the proportional contrast differ between the regimes, confirming distinct simulations.)

| city    | regime   | contrast                       | n_days | mean_daily_diff | ols_slope_per_day | newey_west_t | t |
|---------|----------|--------------------------------|--------|-----------------|-------------------|--------------|---|
| austin  | scarce   | full_vs_median_arm             | 276    | -578.855        | -4.558            | -15.420      |   |
| austin  | scarce   | full_vs_implied_mean_arm       | 276    | -714.163        | -5.451            | -12.386      |   |
| austin  | scarce   | valbest_vs_naive_greedy        | 276    | -1810.605       | -12.512           | -18.676      |   |
| austin  | scarce   | greedy_vs_proportional_valbest | 276    | 687.572         | 6.381             | 8.079        |   |
| austin  | scarce   | LEVEL_proportional_valbest     | 276    | 59359.065       | 393.880           | 34.061       |   |
| austin  | moderate | full_vs_median_arm             | 276    | -676.362        | -5.492            | -12.943      |   |
| austin  | moderate | full_vs_implied_mean_arm       | 276    | -875.678        | -7.048            | -10.288      |   |
| austin  | moderate | valbest_vs_naive_greedy        | 276    | -2637.562       | -20.668           | -56.620      |   |
| austin  | moderate | greedy_vs_proportional_valbest | 276    | 789.036         | 7.421             | 8.257        |   |
| austin  | moderate | LEVEL_proportional_valbest     | 276    | 38592.065       | 243.881           | 21.090       |   |
| austin  | generous | full_vs_median_arm             | 276    | -872.409        | -6.736            | -15.431      |   |
| austin  | generous | full_vs_implied_mean_arm       | 276    | -1047.486       | -8.630            | -12.770      |   |
| austin  | generous | valbest_vs_naive_greedy        | 276    | -5412.554       | -37.550           | -36.511      |   |
| austin  | generous | greedy_vs_proportional_valbest | 276    | 895.217         | 8.550             | 8.611        |   |
| austin  | generous | LEVEL_proportional_valbest     | 276    | 24777.004       | 143.882           | 12.442       |   |
| chicago | scarce   | full_vs_median_arm             | 276    | -3434.554       | -18.827           | -12.345      |   |
| chicago | scarce   | full_vs_implied_mean_arm       | 276    | -4633.688       | -26.374           | -13.059      |   |
| chicago | scarce   | valbest_vs_naive_greedy        | 276    | -9744.841       | -60.852           | -28.347      |   |
| chicago | scarce   | greedy_vs_proportional_valbest | 276    | 3402.880        | 19.789            | 14.643       |   |
| chicago | scarce   | LEVEL_proportional_valbest     | 276    | 134647.402      | 745.261           | 10.606       |   |
| chicago | moderate | full_vs_median_arm             | 276    | -4887.888       | -34.848           | -49.498      |   |
| chicago | moderate | full_vs_implied_mean_arm       | 276    | -6076.783       | -43.377           | -43.385      |   |
| chicago | moderate | valbest_vs_naive_greedy        | 276    | -14018.500      | -99.213           | -32.959      |   |
| chicago | moderate | greedy_vs_proportional_valbest | 276    | 5205.790        | 38.566            | 43.876       |   |
| chicago | moderate | LEVEL_proportional_valbest     | 276    | 72340.402       | 295.262           | 4.202        |   |
| chicago | generous | full_vs_median_arm             | 276    | -4911.076       | -12.285           | -0.886       |   |
| chicago | generous | full_vs_implied_mean_arm       | 276    | -6047.188       | -16.538           | -0.988       |   |
| chicago | generous | valbest_vs_naive_greedy        | 276    | -20437.344      | -130.323          | -9.853       |   |
| chicago | generous | greedy_vs_proportional_valbest | 276    | 5324.014        | 13.986            | 0.917        |   |
| chicago | generous | LEVEL_proportional_valbest     | 276    | 19751.373       | -51.470           | -0.976       |   |
| nyc     | scarce   | full_vs_median_arm             | 276    | -29346.533      | -213.706          | -87.992      |   |
| nyc     | scarce   | full_vs_implied_mean_arm       | 276    | -34431.540      | -246.049          | -50.258      |   |
| nyc     | scarce   | valbest_vs_naive_greedy        | 276    | -54137.612      | -382.018          | -55.941      |   |
| nyc     | scarce   | greedy_vs_proportional_valbest | 276    | 30481.504       | 220.042           | 62.179       |   |
| nyc     | scarce   | LEVEL_proportional_valbest     | 276    | 529074.931      | 3961.562          | 64.340       |   |
| nyc     | moderate | full_vs_median_arm             | 276    | -36728.659      | -286.308          | -75.131      |   |
| nyc     | moderate | full_vs_implied_mean_arm       | 276    | -42909.221      | -326.722          | -91.909      |   |
| nyc     | moderate | valbest_vs_naive_greedy        | 276    | -68753.630      | -511.885          | -146.544     |   |
| nyc     | moderate | greedy_vs_proportional_valbest | 276    | 38156.899       | 297.572           | 72.649       |   |
| nyc     | moderate | LEVEL_proportional_valbest     | 276    | 293624.931      | 2261.562          | 36.730       |   |
| nyc     | generous | full_vs_median_arm             | 276    | -35670.877      | -308.964          | -30.561      |   |
| nyc     | generous | full_vs_implied_mean_arm       | 276    | -41400.859      | -350.388          | -38.135      |   |
| nyc     | generous | valbest_vs_naive_greedy        | 276    | -71800.659      | -558.941          | -80.896      |   |
| nyc     | generous | greedy_vs_proportional_valbest | 276    | 37197.471       | 329.065           | 26.151       |   |
| nyc     | generous | LEVEL_proportional_valbest     | 276    | 63800.714       | 566.092           | 9.329        |   |
| sf      | scarce   | full_vs_median_arm             | 275    | -2518.247       | -16.359           | -20.023      |   |
| sf      | scarce   | full_vs_implied_mean_arm       | 275    | -2658.233       | -17.806           | -24.147      |   |
| sf      | scarce   | valbest_vs_naive_greedy        | 275    | -4528.535       | -31.304           | -42.528      |   |
| sf      | scarce   | greedy_vs_proportional_valbest | 275    | 2772.956        | 17.481            | 16.062       |   |
| sf      | scarce   | LEVEL_proportional_valbest     | 275    | 107245.262      | 871.443           | 40.272       |   |
| sf      | moderate | full_vs_median_arm             | 275    | -2518.247       | -16.359           | -20.023      |   |
| sf      | moderate | full_vs_implied_mean_arm       | 275    | -2658.233       | -17.806           | -24.147      |   |
| sf      | moderate | valbest_vs_naive_greedy        | 275    | -4528.535       | -31.304           | -42.528      |   |
| sf      | moderate | greedy_vs_proportional_valbest | 275    | 2755.022        | 17.480            | 16.066       |   |
| sf      | moderate | LEVEL_proportional_valbest     | 275    | 58963.196       | 521.445           | 24.098       |   |
| sf      | generous | full_vs_median_arm             | 275    | -969.404        | -10.830           | -9.321       |   |
| sf      | generous | full_vs_implied_mean_arm       | 275    | -968.091        | -11.890           | -7.966       |   |
| sf      | generous | valbest_vs_naive_greedy        | 275    | -2467.349       | -25.675           | -15.633      |   |
| sf      | generous | greedy_vs_proportional_valbest | 275    | 833.004         | 10.344            | 7.062        |   |

## 8 Computational Resources

All experiments run on a single Linux container (no GPU). Measured wall-clock times for the corrected rerun: 311 acquisition 29–54 minutes per run on hosted CI runners (portal-side query latency); weather acquisition under 2 minutes; panel and feature construction under 1 minute; the forecasting stage (404 evaluations including rolling-origin folds, stage-censored pooled models, the dual-fit quantile model, and censored zero-shot transfer) approximately 13 minutes; the decision stage (324 primary simulations, 1,125 sensitivity simulations, selection experiment, and 144 block-bootstrap intervals) approximately 9 minutes; inference and artifact generation under 2 minutes. Python 3.11; package versions pinned in `reproducibility/pip-freeze.txt`; global seed 20260609; the decision layer is Monte-Carlo-free.

## 9 Ethics and Responsible-AI Statement

**Data.** All inputs are public administrative aggregates (daily counts of service requests by category) and public weather observations. No individual-level, household-level, or address-level information is acquired, stored, or processed; the acquisition layer requests server-side aggregation precisely so that record-level attributes (including free-text descriptions and locations) never enter the pipeline.

**People inside the data.** Each request count summarizes residents’ attempts to obtain public services. Two ethical asymmetries follow. First, reported volume under-represents communities with lower propensity or ability to report (Kontokosta et al., 2017; Kontokosta and Hong, 2021); optimizing service against reported volume can therefore divert resources from quiet need toward vocal demand. This study’s city-level action space cannot reallocate across neighbourhoods, which contains but does not eliminate the concern: the family-level analogue (e.g., systematically deprioritizing housing complaints relative to noise) is measured and reported per policy. Second, the priority weights are normative inputs; we publish them as configuration, sweep them, and explicitly assign their real-world selection to accountable human decision-makers.

**Intended and unintended use.** The system evaluated here is a research benchmark. It is not a deployable dispatch tool, has no city involvement or endorsement, and the manuscript states this in every section where simulation results appear. Misuse risk (treating simulated capacity numbers as operational guidance) is mitigated by classifying all operational parameters as sensitivity-only and reporting ranges rather than recommendations.

**Human oversight.** The intended human-AI division of labour — forecasts and allocation suggestions with uncertainty made visible, final allocation authority retained by planners — is described in the discussion and is a design commitment of the broader research programme, not an evaluated capability of this study.

## 10 Reproducibility Instructions

**One-command rebuild.** From the repository root:

```

pip install -r reproducibility/pip-freeze.txt # AUTHORITATIVE frozen
# environment (Python 3.11.15, pandas 3.0.3);
# requirements.txt is a broad development file only
make all # panel -> features -> forecasts -> simulated
# decisions -> inference -> artifacts
python3 scripts/run_diagnostics.py # trend diagnostic (tab10)
python3 -m pytest tests -q # full guard suite (37 tests) must pass

```

The guard suite enforces the frozen protocol: training-only budget calibration invariant to test perturbation; validation-only selection invariant to corrupted test metrics; recomputed censor dates on every transfer row plus a fit-time proof manifest for pooled-model stage censoring (Amendment C11); a single fitted model behind all uncertainty arms; structural absence never encoded as zero; exact budget conservation; forbidden-terminology scans; configuration–output agreement; source-identifier consistency; and provenance hashes binding every artifact to its generating run.

**Determinism.** A single global seed (20260609) controls every stochastic component (NumPy, scikit-learn, LightGBM); expected-value allocation uses a deterministic 99-point inverse-CDF grid rather than Monte Carlo, so decision results are bit-stable given forecasts.

**Data provenance.** Raw acquired layers are versioned in `data/raw/` together with JSON manifests recording: source authority, dataset id and page, license note, every query URL issued, retrieval timestamps (UTC), uncompressed and compressed SHA-256 checksums, row counts, and — for 311 sources — the portal’s own dataset metadata (`rowsUpdatedAt`, column inventory) at retrieval time. Re-running `make acquire-311 acquire-weather` re-derives the same layers from the sources (counts for recent periods may drift as portals backfill; the manifests pin what this paper used).

**Environment.** Python 3.11; exact package versions in `reproducibility/pip-freeze.txt`; no GPU required; total pipeline runtime and per-stage costs in Section 8 of this supplement.

**Acquisition environment note.** The research sandbox used for this project blocks direct access to the four city portals; the 311 acquisition therefore ran through the repository’s GitHub Actions workflow (`.github/workflows/acquire-311-data.yml`) on GitHub-hosted runners, executing the same committed script and pushing the raw aggregates and manifests back to the branch. The first attempt with full-window queries timed out against the NYC portal and was replaced by per-calendar-year chunking; both attempts are preserved in the repository’s Actions history.

## References

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